BENGUEDDA LAUZON Elias

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Seeking a 6-month Trading / Quant Research internship (gap year) from January 2027

Professional Experience

Société Générale – Pricing Team, Paris **Jun 2026 – Dec 2026**
Quantitative Research Intern – Derivatives Pricing

- Priced callable bonds under Hull–White 1F: Jamshidian decomposition (European), Monte Carlo + Longstaff–Schwartz (Bermudan, American)
- Built and backtested correlation models (EWMA, DCC-GARCH, Ornstein–Uhlenbeck) to price Worst-of/Best-of; EWMA adopted by the desk

Natixis Trading Solutions, Paris **Summer 2025**
Quantitative Intern – Equity & FICC

- Built automated Python pipelines for daily P&L monitoring and position tracking on the trading floor
- Developed performance analysis tools on equity derivatives data

Quantitative Projects

All papers & code on **portfolio-eliasbenguedda.fr**

Heston Stochastic Volatility Calibration – *CAC40 options, Python, SciPy*

- Calibrated **Heston** to 142 CAC40 options via semi-closed Fourier formula (Lewis, Carr–Madan FFT, COS): IV RMSE 0.42
- Bootstrapped the forward and dividend curve from put–call parity; time-dependent extension: RMSE 0.12

Yield Curve Construction – *€STR, EURIBOR, US Treasuries*

- Bootstrapped OIS and dual-curve EURIBOR curves, validated against QuantLib; 7 interpolation methods compared (monotone convex, Nelson–Siegel–Svensson)
- **PCA** on 181 Treasury curves (level, slope, curvature) and key-rate DV01 risk

Autocallable Notes Pricer – *Athena, Phoenix, Monte Carlo*

- Monte Carlo pricer under calibrated Heston with variance reduction (antithetic, control variate), Brownian bridge
- Fair coupons, scenario statistics and Greeks (vega, dividends) of 35 product variants

Education

Mines Saint-Étienne (Grande École d'ingénieurs), France **2024 – 2028**
MEng Applied Mathematics & Data Science (gap year 2026–2027)

Probability, stochastic processes, optimization, machine learning, statistical modelling

Aix-Marseille University, France **2025 – 2026**
Master in Fundamental Mathematics

Stochastic calculus, PDEs, numerical methods, measure theory

École Normale Supérieure de Lyon (ENS), France **2025 – 2026**
Advanced Courses in Machine Learning & Stochastic Modelling

Reinforcement learning, probabilistic modelling, convex optimization, statistical learning theory

Lycée Janson de Sailly, Paris **2022 – 2024**
Classes Préparatoires MPSI/MP (Mathematics & Physics)

Intensive training in mathematics, physics and computer science for the Grandes Écoles entrance exams

Skills & Interests

- **Programming:** Python (NumPy, SciPy, Pandas, PyTorch, scikit-learn), QuantLib, SQL, Git, LaTeX, C# (basics), C++ (basics)
- **Tools:** Numerix, Bloomberg
- **Languages:** French (Native), English (Fluent), Spanish (Basic)
- **Interests:** Karting (French National Championship), Aviation (student pilot)